

Guoqiang Luo

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Native Mandarin Chinese · Fluent English (technical reading & writing)

SUMMARY

Materials Science and Engineering graduate working on data-driven property prediction and inverse design for cast aluminium alloys, focusing on the A356 (Al-Si-Mg) and A354 (Al-Si-Cu-Mg) systems: UTS and elongation (EL) prediction from composition and process parameters, physics-informed feature selection, applicability domain analysis, and time-blind validation design. Undergraduate research background in A356 welding metallurgy, with hands-on experience in optical metallography, EBSD, intergranular corrosion testing, and mechanical property evaluation. Available approximately 15 hours per week for remote, long-term work.

CORE COMPETENCIES

Cast Aluminium Alloys	A356 (Al-Si-Mg) and A354 (Al-Si-Cu-Mg); composition–property relationships; heat treatment and temper condition effects; eutectic Si morphology
Property Prediction	Ultimate tensile strength (UTS) and elongation (EL) prediction from composition and process parameters; gradient-boosting models with per-alloy models; physics-informed feature selection by strengthening mechanism
Model Validation	Time-blind holdout design; cross-validation; missing-data imputation evaluation; A/B testing of modelling choices; applicability domain assessment
Inverse Design	Multi-objective design of composition and process windows for target strength–ductility combinations
Microstructural Analysis	Optical metallography; EBSD grain size analysis; eutectic Si morphology and grain structure
Welding & Corrosion	Welding of cast Al-Si alloys; weld metal, fusion line, heat-affected zone (HAZ), and base metal characterization; intergranular corrosion testing and corrosion depth measurement
Mechanical Testing	Tensile testing (UTS / EL) and hardness testing
Technical Translation	Chinese ↔ English scientific and technical documents; terminology consistency; standards and patent abstracts

EDUCATION

MSc, Materials Science and Engineering — Central South University, Changsha, China	2025 – present
BSc, Materials Science and Engineering — Central South University, Changsha, China	2021 – 2025

RESEARCH EXPERIENCE

Central South University — Property prediction and inverse design for A356 and A354 cast aluminium alloys	2025 – present
- Predicting ultimate tensile strength (UTS) and elongation (EL) for A356 (Al-Si-Mg) and A354 (Al-Si-Cu-Mg) cast aluminium alloys from composition and process parameters. - Gradient-boosting models with per-alloy models; physics-informed feature selection grouped by strengthening mechanism with collinearity control. - Designed the validation scheme: time-blind holdout, cross-validation, missing-data imputation evaluation, and A/B testing of modelling choices. - Applicability domain analysis to flag predictions falling outside the training data range. - Multi-objective inverse design to identify composition and process windows meeting target UTS / EL combinations.	

UNDERGRADUATE RESEARCH

Welding of A356 cast aluminium alloy — Central South University	2024 – 2025
- Investigated how welding affects the microstructure, mechanical properties, and intergranular corrosion resistance of A356 cast aluminium alloy. - Characterized the welded joint by optical metallography, examining the weld metal, fusion line, heat-affected zone, and base metal; quantified grain size across the weld zones by EBSD. - Evaluated intergranular corrosion resistance by corrosion depth measurement, and related it to the observed microstructure. - Measured the mechanical properties of the welded joint and compared them with those of the base metal.	

LANGUAGES & AVAILABILITY

Chinese (Mandarin)	Native
English	Fluent reading and writing (technical documentation, standards, journal articles); conversational speaking
Availability	Approximately 15 hours per week · UTC+8 · Long-term remote work